

Carpathian OER Commons — Compendium

Philosophy, Pathway, and Structure in One Document

Michel Garand | 2026-09-29 | Version v0.3 | CC BY-SA 4.0

The compendium of the Carpathian OER Commons: one document that says why the commons exists and what it stands on, how the guide is built and how it travels, how the settlement grows and governs itself, and how the project, its repositories, and its documents are organised. It gathers three papers — the foundations paper, EN v0.3; the pathway paper, EN v0.4; and the open hamlet paper, EN v0.1 — which stay published beside it, and adds an overview of the project, its structure, and its documents (Part 0). Section 33 says what changed in the gathering.

v0.2, 2026-09-28: the market garden enters the map. The kitchen garden row is replaced by a market garden and a food storage row in Land, forest, food, and a growers' and eaters' association joins Settlement — forty-two modules. The compost row becomes the garden stream, and the terra preta row carries the whole toilet stream, and the compost module is published at L0 as a marked draft. Sections 1, 3, 4, 14, 15, 16, 18, 25, 26.2, 26.5, 31, 33, and 34 changed; nothing else.

v0.3, 2026-09-29: the toilet stream decided as one stream; the terra preta and biochar modules published at L1, converted from the sanitation paper, and the market garden at v0.4. Sections 3, 4, 31, 33, and 34 changed; nothing else.

Section 2 governs everything here and everything the commons publishes. No child appears in any form; the programme is described in standing format only. The site is not named, and nothing here places it; every site plan is schematic — relationships only, no distances, no terrain, no names. Every idea taken from elsewhere is named with its source and with what we change. Every point of Hutsul practice is marked [for the elders]: none of it is ours to state until they have confirmed, corrected, or withheld it.

A working draft, to be changed in the open by the people it describes and the people who use it. Nothing in it is carved in stone.

Part 0 — The commons at a glance

1. What the commons is

The Carpathian OER Commons is an open guide to how one place in the Hutsul highlands of the Ukrainian Carpathians is built and run: a residential forest school for children and youth, a place of retreat and continuing education for adults, and a small settlement growing around them — a hamlet, *присілок*, the working term until the elders give theirs [for the elders].

The guide is written as modules, forty-two of them, each with the same shape and each carrying a maturity level that states what may be claimed from it. Beside the modules it carries the place's own pattern language, discovered and not imposed. Today two modules stand at a level the site's own experience backs, fifteen are at design stage, two — the market garden and the compost — are published at L0 as marked drafts, and twenty-three are empty rows, left visibly empty.

The initiative. The commons grows out of Soil and Peace, a co-created community regeneration practice based on one conviction: that the hand that tends soil and the hand that tends what lies between people are the same hand. The project uses, and helps to develop, the framework of the Ubuntu Bioregional Economic Commons (UBEC) DAO, and draws on the Erdpuls open educational resources as worked examples. Section 6 says more.

Open educational resources. Documents are released under CC BY-SA 4.0, drawings under a hardware licence still to be chosen, and measurement data openly, with attribution only. That places the guide inside the definition UNESCO adopted in 2019: "learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others." The licence makes the guide open. It does not by itself make it a commons.

Why a commons. In Elinor Ostrom's work a commons is a shared resource held by the people who depend on it, under rules they make and can change (Governing the Commons, 1990). The resources she studied — pastures, fisheries, irrigation — are used up by use: what one takes, another cannot. A guide is not; reading it takes nothing from it. What can be used up here is different: the trust of the household the guide describes, the consent of the elders whose knowledge it draws on, the safety of a place in a country at war, and the honesty of the record. Those are what this commons has to govern. Knowledge treated as a commons is the subject of later work by Charlotte Hess and Elinor Ostrom (Understanding Knowledge as a Commons, 2007), and that is the reference when governance is built — in Phase 5 of the pathway (Section 17), and not before. A commons with one member needs no constitution.

The measure. Borrowed from Open Source Ecology and narrowed (Section 9.1): **the guide is complete when another site has built from it, in another valley, and their figures sit in it beside ours.** Not when every module is written, not when it is translated into six languages, not when it is published. A guide with six modules and one honest replication is further along than thirty modules read by nobody. It is the only measure this document uses for the commons.

What changes in the people who take part is not measured by the guide and is not published. The programme's qualified staff hold that, inside the programme. The commons is measured by whether what it describes can be built again elsewhere.

2. Safeguarding governs everything here

The guide is a published artefact, so the project's child-protection rules apply to every line of it, and the aggregation test applies across documents: a drawing in one, a photograph in another, and a date elsewhere can together place a site that no single document placed. When in doubt, the hold closes.

- **No child in any module.** Not in a photograph, not in a caption, not in an example, not in a headcount, not as texture. Programme modules describe standing format only. Children co-create in the living practice; in the record they appear only as the cohort, never as persons.

- **Publish the method, not the place.** Site plans in a published document are schematic — relationships only, without distances, names, or the terrain that identifies a valley, until the decision on naming the site. The site is the hamlet: dispersed homesteads on a slope, drawn with distances, are the most identifying artefact there is, and would place a valley. Construction drawings of a single structure keep their dimensions.
- **Photographs** show structures and systems: no faces, no signage, no distinctive ridgelines, no metadata.
- **Data** leaves the site only at valley resolution: no coordinates, no dates tied to a session, no account belonging to or naming a child (Section 9.5).
- **Regulatory and application material** that names an entity travels to funders, not to the repository.
- **Pattern cards** stay in the workshop. Only the yearly narrative of the place's pattern language is published, at valley resolution.
- **The elders and the households.** Nothing of the elders' is published without their consent, on their terms. Nothing of a resident household's life is published without that household's own Gate 1.
- **Gate 1 throughout.** The household decides what of its own life is described, module by module.
- **Wartime review before every release**, not once at the start. What was safe to publish in one season may not be in the next.
- **A line held beyond the rules.** Children, wartime, a named doorstep, a fixed week, and family circumstance, in one piece, are never published, whatever the rules at the time allow.

The guide is publishable because infrastructure is publishable. The programme modules are the difficult ones, and the right instinct there is to publish the architecture of the safeguarding and very little of the programme's daily life.

For a replicating site, which will not see the project's working rules: Tier 1 — no child named, dated, counted, or placed, ever; Tier 2 — anything more only through consent gates; Gate 1 — each household decides what of its own life is described.

3. How the work is organised

Two repositories, one direction. The workshop is a private repository on GitHub, under `ubecommon`. Everything is written, checked, and argued over there — drafts, sources, L0 and L1 work, open decisions. The publication is a collection, `Carpathian_OER_Commons`, in the public repository `ubecommon.github.io`, beside the `Erdpuls` collection `Pattern Language of Place`, and served by the same machinery: folder pages with a language switcher, a viewer for the markdown, deployment on every push. Material moves one way only, workshop to publication, and nothing is written directly in the publication. Every clone is a full local copy, for when the connection fails, and the hosting costs the household nothing.

Why the release step is a gate. The publication repository is public, and git keeps everything ever committed to it: a file copied there by mistake is public for good, even if deleted the next minute. So nothing is released by hand. A release is a dated list of files; a script copies only what is on the list, refuses any module below L1 that is not a marked draft, any Gate 1 or wartime review line left unfilled, and a foundations document without both, and checks every file against a list of place names kept outside both repositories. The printable PDFs are built after the copy and checked against the same list.

The layout. The same in both repositories, minus what never leaves the workshop.

- **foundations/** — this document and, in grounding/, the three grounding documents (Section 4), by language: foundations/EN/, with UK/ and DE/ beside it.
- **modules/** — the guide itself, language first, then group: modules/EN/utilities/, with UK/ and DE/ beside it; on the public site, a pdf/ folder in each language.
- **standards/** — the module template and the pattern card template, published; the repository conventions, the site parameters, the checker, and the release script, in the workshop only.
- **drawings/** and **data/** — planned: source drawings with their exports, and measurement series, by group. They carry no language.
- **patterns/** — planned, in the workshop: the place's pattern cards, which stay there for good, and the yearly narrative drawn from them, which is released.
- **releases/** and **sources/** — in the workshop only: the release lists, and the three working-paper guides the modules are converted from.

Toolchain. The guide is written in the same markdown standard as everything else in this project, the Erdpuls standard, and checked by the guide's own checker, standards/check_oer_standard.py, before every commit. Other writing about the place stays where it is; the guide is a separate thing with a separate rhythm, because testimony and technical documentation age differently.

- **Text:** markdown, one file per module, checker before commit; file names carry the language and the version, as module_utilities_water_EN_v0_2.md.
- **Drawings:** SVG or FreeCAD, source files committed beside the exports. A PDF of a drawing is not a drawing.
- **Measurements:** plain CSV, one series per file, with a header stating instrument, place at valley resolution, and who took it.
- **Photographs:** metadata stripped, filenames neutral, reviewed against Section 2 before they go anywhere near a server.
- **Figures shared between modules** — capacity, litres per person, the exchange rate — are declared once, in a site parameters file, and cited by name. A change there is carried through every module that cites it in the same commit. Whether each value may be published is decided value by value.
- **Releases:** a dated snapshot when a group of modules reaches L2, so that a replicating site can cite a fixed version.

Languages. English is canonical and settles first; other languages are siblings with a stated revision date, never blocking publication. Ukrainian follows close behind and is corrected over time rather than held back. German is published beside them. The Ukrainian and German texts now published are machine translations, marked as such and provisional until a native reader has reviewed them; where they differ, the English holds. Polish, Romanian, Slovak, and Hungarian come only when a partner in that country takes on the translation as their own work — a machine-translated module with no native reader is worse than none, because it looks checked. The Ukrainian and German machine translations, decided 2026-09-28, are the stated exception: marked provisional on their face, with the English holding, until a native reader has reviewed them.

Versions. A document carries its version in its name and in its front matter. A change of content steps the version; the older file leaves the public site, and the workshop keeps its history.

Where the work stands, 2026-09-29. Phase 0 is largely done: the module template, the pattern card template, the repository conventions, the site parameters, the checker, and the release script. Of the three working-paper guides, water is converted into its module, at L1, and published in English, Ukrainian, and German; sanitation is converted in part — its terra preta and biochar rows are modules at L1, published in the same three languages, with their PDFs, while its toilets, greywater, washroom, and wood heat rows are not; energy is not yet converted. The market garden, the first module of the open hamlet, is published at L0 as a marked draft, v0.4, in English, Ukrainian, and German, with its PDFs; the compost module, which feeds it, is published beside it at L0 as a marked draft, v0.1, in the same three languages, with its PDFs. Phase 1 has not begun: the baseline has not been walked, and the pattern record has not been started.

4. The documents

What each document is, where it lives, and what it governs. Everything published is a working draft, open to co-creation.

What governs.

- **This compendium** — foundations/EN/, published. It governs the commons, and Section 2 governs everything else in it. The foundations, pathway, and open hamlet papers stay published beside it, in foundations/EN/, with the Ukrainian and German beside them. Where the compendium and one of them differ, the compendium holds, and the paper is brought into line in its next version.
- **The repository conventions** — standards/, workshop only. They carry Section 3 into practice: layout, naming, versions, languages, releases. Where they and this document differ, this document holds, and the conventions are corrected.
- **The project prompt** — workshop only: the working instructions for the AI assistant used in drafting, with its session prompts. It follows this document.

The grounding documents. They speak from conviction; the guide speaks from measurement, and neither may borrow the other's authority (Section 7). Published in foundations/EN/grounding/, with the Ukrainian and German beside them.

- **The Eco-Spiritual Philosophy of the Living Forest** — v0.9, by Michel Garand with the host household. The forest school's philosophy: six tenets, from the forest as a living temple without walls or dogmas to the transformation of pain through song and dance; the wheel of the year; and the step from philosophy to programme. Its Hutsul heart comes from the household's own vision for the school; its learning architecture from Soil and Peace and the Pattern Language of Place work.
- **The Forest in the World's Wisdom** — v0.7. A comparative survey of spiritual traditions the practice here learns from, from the Hutsul and Ukrainian folk worldview and the valley's churches to the peoples of the Urals and Siberia, named by their own lands, and traditions of other continents. Background reading for adults: no practice of another people is done with children. Source-checked and stress-tested on 2026-09-27; it records what the check found.
- **Forest School — From Disconnection to Regeneration** — v0.28. The programme proposal: its thesis, the method of a pattern language of place, the programme's components, its pedagogy and safeguarding, and the signs by which the work would show it is working. Written as an application draft, in standing format.

The guide.

- **Modules** — modules/, by language and group. Published: Water: spring, cistern, treatment, v0.2, at L1, designed and costed from estimates, untested; Market garden: beds, rotation, the season, v0.4, at L0 as a marked draft, sized for about 3,000 m² and not fully costed; Compost: the garden stream to the beds, v0.1, at L0 as a marked draft, sized from the market garden's need, with the bought-in share priced; Terra preta: fermentation to soil, v0.1, at L1, designed and costed from estimates, untested, and not to be built before the sanitary service's written answer; Biochar: kiln, quality, storage, v0.1, at L1, designed and costed from estimates, untested. The other thirty-seven are listed in Section 16.
- **The module template and the pattern card template** — standards/, published in English, Ukrainian, and German, so that a module or a card may be written in any language.
- **The site parameters** — standards/, workshop only (Section 3).

The workshop's own. The release lists; the working-paper guides for water, energy, and sanitation, kept unchanged and never published; the pattern cards, once begun.

Not in either repository. Application and funder material naming an entity; photographs until reviewed; other writing about the place.

5. How to read this document

- **The household reads it first.** It describes the place the household lives in and the practice it shares, and it passes the household's Gate 1 before anyone else reads it.
- **It is written for a replicating site.** Sections 2, 8, 9, 11, and 20 are what travels. A site that reads them and takes two modules has used the commons as intended. Part III is for a site that is also growing a settlement.
- **A reviewer or a funder** reads Sections 1, 7, and 12: what is claimed, and what is not.
- **The elders** are not sent a document. What this document marks for them is taken to them in conversation, and their answers change the text. Section 29 gathers the questions.
- **A co-creator** — anyone who takes part — starts with Parts 0 and I, and finds their own work in the modules and in the pattern record.

Part I says why the commons exists and what it stands on. Part II says how the guide is written and how it travels. Part III says how the settlement grows and governs itself. Part IV holds the decisions, the reviews, and the sources. The compendium is both a standalone text and the preface to the guide.

Part I — Philosophy: why the commons exists and what it stands on

6. Background

The initiative. The place the guide describes is where the conviction of Soil and Peace (Section 1) is being worked out, on the ground, with a household and its neighbours.

Peace as infrastructure. Soil and Peace takes the phrase from peacebuilding, where an infrastructure for peace means the relationships and structures that carry a peace beyond its agreements, and reads it at the scale of a household and its ground. John Paul Lederach proposed the idea in *Building Peace* (1997), out of mediation work in Nicaragua. In his words, "we are not merely interested in 'ending' something that is not desired. We are oriented toward the building of relationships that in their totality form new patterns, processes, and structures." Our reading changes the scale and the material — a spring, a forest, a record, a working relationship between neighbours — and keeps his point: the aim is not only to end what is unwanted but to build what holds.

The guide is the checkable case. That peace grows from tended ground is a conviction, and this document does not claim it as a finding. One part of it can be checked. Documentation that others build from, and that outlasts the people who wrote it, is infrastructure in the plainest sense. If a second site builds from the guide and its figures come back into it, that much will have been shown — for the guide, not for peace.

Why a guide. The place is being built in wartime, in mountains that run through seven states: Ukraine, Romania, Poland, Slovakia, Hungary, Serbia, and Czechia. Other initiatives in the range face the same questions — water, power, sanitation, winter, safeguarding, how to hold land — with the same scarcity of money and time. Many published projects describe what worked and leave out what failed and what it cost. The guide exists so that the next site does not repeat the expensive mistakes, and so that what was found here is offered as method, never as a template.

What the war changes. The war is the ground the work stands on, not its subject. It sets hard limits on what can be published (Section 2): no child in any form, no site named while the decision on naming it is open, no plan that places a valley, no data finer than valley resolution, and a wartime review before every release. Those limits are not an inconvenience to be engineered around. They are part of the method, and a site in peacetime may keep them for its own reasons.

7. What the commons stands on

Every idea in this document grew somewhere else first. What we stand on was laid down by others, the way soil is — in layers, from what lived before: a practice of tending ground, a way of documenting that others can build from, a way of looking that lets a place speak first, words lent by peoples who were here long before this work. We receive them as the ground receives what falls on it — slowly, changed by what they meet, until they feed something that was not there before. We take nothing whole. We take as a root takes: only what it can turn into its own growth, and only where it stands. What is ours is not the ideas but the ground they have fallen on — and the honesty to name where each one grew.

The documents this one draws on speak in two registers. The grounding documents — the philosophy of the living forest, the comparative survey, the programme proposal (Section 4) — speak from conviction. The guide speaks from measurement. The commons needs both, and neither may borrow the other's authority.

Held as ground. Four convictions, stated as convictions:

- **Care circulates.** We hold that where work is done together, care runs in every direction until no one is only a giver or only a receiver. The guide's forms of it are concrete: joint attribution, the household's time counted as contribution, cultural work to be funded by gift. The guide does not say that care is the only currency. Every module carries a cost line in hryvnia, and it would be untrue.
- **Land and people are regenerated together.** The work on the ground and the work between people are one work, seen from two sides. The guide measures the ground side — a spring's yield, seedling survival, returning species — and says nothing about the other.
- **Peace is grown as infrastructure.** Section 6.
- **A place teaches, when it is attended to.** Every place has its own pattern language; the task is to notice it, not to invent it (Section 9.2).

Not claimed. The guide does not claim that the forest heals, that the method changes the people who take part, or that the place has become what it hopes to be. Those are the reasons the work is done. They are not results the work has shown, and no module is written as if they were.

The bridge. One practice already carries both registers at once: the dialogue between the body and the instrument. A hand in the soil says sheltered here, exposed there; a thermometer gives a number; neither replaces the other, and the gap between them is where the learning is. The commons takes that as its discipline for everything it writes: a conviction may say what to look at; only a measurement may say what was found.

8. Principles

8.1 *Not one to one*

No language, method, or tradition is taken over whole — not Alexander's, not Steiner's, not the Erdpuls method's, not any people's. And none is offered whole: a site that builds from the guide takes modules, not the model, and grows its own form from its own ground. The rule runs both ways, and that is its point.

Its source is Alexander's caveat on his own book: "each pattern may be looked upon as a hypothesis, like one of the hypotheses of science" (A Pattern Language, 1977, p. xv). What we change is its direction. He applied it to what he offered; we apply it also to what we take.

8.2 *With, not for*

MIT, nicht FÜR — the ethical spine of Soil and Peace, kept in its German: building with a place and its people rather than for them. In the guide it governs three relations. Between the author and the place: the guide describes our own work, never a place's life on its behalf. Between the guide and its readers: replication is an invitation, never an export. Between the work and those who take part: everyone is a co-creator (Section 8.4). Work that begins as done with people and lasts long enough becomes work they lead; that is what "with" means over time, not a third term.

8.3 *Honest before useful*

A module that says "we have not tested this" is worth more than one that reads well. Every module states its level at the top, from L0, an idea, to L5, improved by others and merged back, and readers act on the level, not on the prose (Section 15). Every figure is measured, quoted, or marked [set] or [assumed]; a cost without a date is a rumour. The failure of open documentation is work at one level published in the language of a higher one. This document is held to the same rule, and Section 12 applies it to its own convictions.

8.4 Co-creators

Everyone who takes part — children and adults alike, the household, the trades, the elders, guests, residents, the people of other sites — is a co-creator. The living-lab literature says users. The Erdpuls method does not, and the commons follows it, because a pattern language that people have not discovered and named themselves is someone else's, imposed. Attribution follows the work: where it was joint, the credit is joint. An open licence that quietly credits shared work to one author is a theft dressed as generosity.

8.5 Not ours to give

Knowledge that belongs to a people or to particular persons is not open because someone wrote it down. Hutsul practice — forms, songs, craft, the knowledge of a valley — enters the guide only with the consent of those whose knowledge it is, on terms they set, and some of it will not enter at all [for the elders]. The default is to describe our own use of a practice and point to its holders, rather than publish the practice as if it were ours to give. Sacred sites, holy ground, and graves are recorded nowhere, in the workshop or in publication, unless the elders ask. The same holds for every tradition named in Section 10: we say whose it is and how we use it, and we do not speak for it.

8.6 Across the borders

The commons is named for the range, not for a country. The frame is bioregional. Peter Berg and Raymond Dasmann introduced the bioregion and reinhabitation as paired terms in "Reinhabiting California" (The Ecologist, 1977), and wrote, in the essay as reprinted the following year, that "the final boundaries of a bioregion are best described by the people who have lived within it, through human recognition of the realities of living-in-place." Here that sentence is a rule: the boundaries of the place, like its names, are for the people who live in it and for the elders to draw [for the elders].

A cross-border frame for the range already exists in the Carpathian Convention. It is worth understanding before inventing a network: it has working groups, funded programmes, and a habit of cooperation that predates this project. Its current standing is to be checked before it is cited in any application.

What travels across the borders: sizing methods, the order of decisions, failure modes and the record of what failed, the maturity discipline, the safeguarding architecture. What does not travel, and must be localised by whoever is there: regulation — sanitary, building, water, child protection, all national; prices and trades; species and planting calendars; the language of the programme; the relationship with a local authority, which is always personal.

Each module therefore carries one regulatory row per country, dated and signed by whoever checked it. An empty row stays visibly empty, so that a reader in Romania can see at a glance that nobody has checked Romania.

9. Where the method comes from

9.1 *Documentation as the deliverable*

Open Source Ecology, founded by Marcin Jakubowski in 2003, sets out to publish open designs for the fifty industrial machines it counts as needed for a small civilisation with modern comforts, the Global Village Construction Set — built and tested on its own farm, designed to be modular, repairable, and made locally at a fraction of commercial cost. Four of its habits are taken.

- **Documentation is the deliverable.** A thing built and not documented has not been finished. OSE treats a demonstrated, replicable production model as the completion metric, not a working prototype.
- **Modularity.** Independent parts that interoperate, so that a reader takes the two modules they need without swallowing the whole.
- **Lifetime design and serviceability.** Built to be repaired by the people who use it, with the documentation that makes repair possible.
- **Replication is the measure of success.** Not how good the original is, but how quickly others can build and improve on it.

Four are not taken.

- **The machines.** This is not a fabrication project. The technologies here are a cistern, a reed bed, a wood stove, a battery, a nursery — mostly older than industry and mostly built by local trades.
- **The completeness claim.** No set of fifty anything. The guide covers what this site actually does, and says plainly where it stops.
- **The rhetoric.** No post-scarcity, no new economy, no civilisation-building language. The register here is a household in a war, describing what it built and what it cost.
- **Open by default, everywhere.** A hardware project can publish everything. This one cannot: children, a wartime location, and knowledge that is not ours to license all set limits (Section 2).

9.2 *Discovery, not imposition*

Christopher Alexander and his colleagues set out 253 patterns in *A Pattern Language* (1977), from the region to the construction detail. In their definition, a pattern "describes a problem which occurs over and over again in our environment, and then describes the core of the solution to that problem." The book offers a language to be applied.

The Erdpuls method departs from him once: it arrives with a way of discovering a place's own pattern language, not with a language to impose. The commons departs a second time. Alexander's patterns are prompts for looking, never a specification (Section 26). A pattern of his that the place does not carry is not a lack; it is a question — what answers that tension here — and may be a sign that the place answers the same tension another way.

The place may also hold forms that are not patterns in his sense at all. A pattern in his book is a context, a tension, and a resolution in space. This landscape may speak in other forms: a rhythm — the herd going up to the polonyna and back; a movement — the path from homestead to spring; a relation held in custom rather than in building. The record stays open to whatever form the place shows.

Three kinds of pattern are kept apart: the place's, named by the household, the co-creators, and the elders; Alexander's, cited by number and name as prompts; and a module's own design rules, which stay inside the module and are tested by its maturity level (Section 27.2).

9.3 Delicate empiricism

"Es gibt eine zarte Empirie, die sich mit dem Gegenstand innigst identisch macht und dadurch zur eigentlichen Theorie wird" — there is a delicate empiricism that makes itself wholly one with its object, and so becomes theory in the true sense (Goethe, *Maximen und Reflexionen*; the translation is ours). A second maxim gives the order of work: no phenomenon explains itself; only many, surveyed together and methodically ordered, give something that may count as theory.

The Erdpuls method takes from Goethe the order of knowing: observation before explanation, the body as the first instrument, simple measurement as the second. Later writers, Henri Bortoft among them, have set Goethe's way of seeing out in stages; those stages are theirs, not Goethe's, and are cited as such where they are used.

9.4 Growing like a plant

Goethe's *Metamorphosis of Plants* (1790) — one form, expressed differently wherever it grows — gives the Erdpuls method its way of looking, and the settlement its way of growing.

- **Roots before shoots.** The place is walked and measured before anything is built — Phase 1 of the pathway.
- **Grow from where it stands.** From the existing buildings, the spring, the slope; build on the poorer ground and keep the best.
- **Grow toward light and water.** Siting follows sun and source, read on the ground, not on a plan.
- **Many small parts, no single centre.** Several small buildings and households rather than one block, so that the loss of one part does not end the whole.
- **Grow in season.** Building, like planting, keeps to its windows.
- **Sense everywhere.** Observation is distributed among everyone who lives, works, and learns here — citizen science as the plant's many small sensing points.
- **Shed and compost what fails.** A failed design is recorded, taken down, and returned to the record as what we got wrong.
- **Seed, not copy.** Another site does not copy this place. It takes the method as a seed and grows its own form from its own ground — which is what L4 means.

Alexander asked the same of a community in *The Oregon Experiment* (1975): piecemeal growth rather than a plan built at once.

This is a metaphor for method, not a claim about plants. The phrase echoes Aldo Leopold's "thinking like a mountain" (*A Sand County Almanac*, 1949). The picture of a living body with no single centre, sensing everywhere, is close to Stefano Mancuso's argument about plant intelligence, which other biologists dispute. Neither is used as evidence.

9.5 Citizen science and the living lab

The term citizen science has two origins, reached independently. Alan Irwin (*Citizen Science*, 1995) meant a science answerable to citizens' concerns and knowledge that citizens produce themselves; Rick Bonney (1996) meant members of the public contributing data to a research project. Here, observation and measurement are done by the people who live, work, and learn at the place, with the body first and simple instruments second, and the practice stands nearer Irwin's sense. What they record is open data, attribution only. The living lab is how the place is run; citizen science is how it comes to know itself.

It has three homes:

- **Modules** — the measure-first discipline: spring yield, soil, the battery through December. The measurement series in the toolchain are citizen science data.
- **Pattern cards** — the Instrumental Data field.
- **The hamlet** — residents and retreat guests keep the long series between sessions.

The valley-resolution rule. It departs from the openness the other sense assumes, and from the Erdpuls method, whose documents provide for contributing freely to open platforms. Data leaves the site only at valley resolution, under the programme's instrument filter: no coordinates, no dates tied to a session, no account belonging to or naming a child. Any contribution to a public platform is made, if at all, by an adult, with the location obscured, after the wartime review. The departure is deliberate, and a replicating site can weigh its two reasons for itself: a country at war, and children who cannot consent to being placed.

Living lab is used as a descriptive frame, not an affiliation. The term is not protected, and no connection is claimed with any outside network or certification body. Its elements are drawn from the living-lab literature, and it means five things together: work in a real setting; co-creation; many parties at the table — household, trades, elders, guests, other sites; many methods; and a cycle of trying, measuring, and revising. The literature speaks of users; the commons keeps the elements and departs from the word (Section 8.4). The maturity levels are the lab's cycle made explicit: L1 to L3 at this site, L4 and L5 across sites.

9.6 Head, heart, and hand

Learning with head, heart, and hand is Johann Heinrich Pestalozzi's (1746–1827). Rudolf Steiner's thinking, feeling, and willing is a later parallel; the Erdpuls method draws on it as a developmental psychology, not as a worldview, and the commons needs only Pestalozzi's form.

The four stages the Erdpuls method uses — Awareness, Acknowledgment, Attitude, Action — are its own. The method is designed and documented in the Erdpuls OER collection, in the Pattern Discovery Toolkit, EN v1.3, and the Pattern Language Assembly Guide, EN v1.2. It is claimed here as designed and documented, and no more.

10. Borrowed words

Each word below belongs to someone. This section says whose, and how the commons holds it.

Toloka. Neighbours doing together a piece of work none could do alone, repaid with a feast. The practice is not only Carpathian: its names run from talka in Latvian and Lithuanian and talgud in Estonian to łoka in Polish, kalaka in Hungarian, and toloka in Ukrainian. Shared work has also been turned into owed work: Lithuanian estate regulations of 1557 already list talka among the duties peasants owed the estate. One study of the Hutsul region records toloka also as family pasture held across generations near the homes — family land, not common land. Which sense this valley uses is for the elders [for the elders]. The guide uses the word for shared work freely given, and never as a name for unpaid labour an economy depends on.

Ubuntu. The Nguni proverb umuntu ngumuntu ngabantu, in Xhosa and Zulu: a person is a person through other persons. "I am, because we are; and since we are, therefore I am" is John Mbiti's rendering (African Religions and Philosophy, 1969), not the proverb's translation. The idea is contested within African philosophy; Mbiti, among others, warns against reading it as a claim that Africans are naturally harmony-seeking. The framework this commons works within, UBEC, the Ubuntu Bioregional Economic Commons DAO, carries the word in its name. We describe our use of it — that personhood here is read as reaching into the land a community belongs to — and say that this extension is ours, not the proverb's. We do not speak for it.

Social threefolding. Rudolf Steiner proposed it in memoranda of 1917 and campaigned for it in 1919: freedom in cultural life, equality in the life of rights, fraternity in economic life. The hamlet's adults use it as a lens (Section 25). It is held as method, not worldview; never taught to children; asked of no family. It is contested, and so is its author: the 1917 memoranda cast Western democracy as Anglo-American domination of central Europe. Communities partly organised on these lines have also shown where it breaks; Section 25 records the caution.

Forest school. The name belongs to an existing movement: settings that emerged in Denmark in the 1950s, reached the United Kingdom through Bridgwater College in 1993, and have had a professional body there, the Forest School Association, since 2012. The commons uses the words descriptively. It claims no affiliation and no accreditation.

Hutsul. The people and the highland whose practice this place stands inside. Nothing Hutsul in the commons is ours: the form of settlement, the enclosed homestead (ґражда) and the outlying group of homesteads (кyт), the polonyna and its season, the local sense of toloka, songs, dances, and the customs of mourning are for the elders to confirm, correct, or withhold [for the elders]. Where this document or the grounding documents draw on them, those passages are marked for the elders and wait on their word; nothing in them is stated as confirmed until the elders have spoken.

The world's traditions. The comparative survey behind the philosophy names traditions from the Baltic to Aotearoa that the practice here learns from. The commons names a tradition by its people, never practises it as its own, and never uses it as evidence. That a shape recurs across many traditions says something about people. It does not show that a method works.

11. What the commons will not do

- Place a child in any form: no name, face, detail, date, or count. The programme is described in standing format only.
 - Name the site, or publish anything that places it — a plan with distances or terrain, a photograph with a face, signage, or a ridgeline, data finer than valley resolution — until the decision on naming the site is taken.
 - Claim above its level, or print a figure that is not measured, quoted, or marked.
 - Publish what the elders have not consented to, or anything of a resident household's life without that household's own Gate 1.
 - Teach a doctrine. The convictions are lived in the open, where a family can see them before choosing; none is taught as truth, and no family's faith is asked to move.
 - Speak for a people or a tradition (Sections 8.5 and 10).
 - Sell. No module is held back to create demand for help. Help beyond the guide is a separate arrangement and does not close the guide.
 - Launch. No network, no campaign, no movement. One other site first.
 - Build an economy on unpaid residents, or on work that is owed rather than given.
 - Carry application or funder material. Anything naming an entity goes to funders, not to the repository.
-

12. Where the convictions stand against the evidence

- **The method travels.** Tested when a site in another valley builds from the modules and returns its figures, which is L4. Today: no such site.
- **The guide stands without its writers.** Tested when a site builds from it with no help from the people who wrote it. Today: untested.
- **Land regenerates with the work.** The nursery and planting modules would measure seedling survival and returning ground flora across seasons. Today: the nursery module designed at L1, planting at L0, no season measured.
- **A place teaches, when it is attended to.** Tested when pattern cards are confirmed by independent observers across groups and seasons. Today: the settlement walk not yet made, the pattern record not yet begun.
- **Care circulates.** The guide measures nothing that would test it and claims nothing from it.
- **Peace grows from tended ground.** Nor this. The guide is one piece of infrastructure the conviction would predict; if it is replicated, that shows the guide works, not that peace grew.

The commons stands on convictions it cannot prove, and on a method it has designed and documented but not yet seen travel. Every module says the same of itself, at its own level.

Part II — Pathway: how the guide is built and how it travels

13. What is being built

Two things are being built at once, and confusing them wastes both.

The first is a place: buildings, water, power, sanitation, a forest, a programme. It exists to serve the children who come and the household that hosts them, and it would be worth doing if no one ever copied it.

The second is a description of the place, accurate enough that people fifty or five hundred kilometres away can build their own version without repeating the expensive mistakes. That description is the guide.

The guide is not a by-product of the first thing. It is its own work, with its own standard, its own labour, and its own cost — and if it is treated as something to write up afterwards, it will not be written. What follows is the pathway to producing it deliberately.

The place is also becoming a settlement. The forest school serves adults too — retreat and continuing education — and around it a small hamlet is planned: households living there, with work, learning, and land held together. The hamlet is part of the place, and its description is part of the guide: the place's own pattern language, discovered and not imposed, and the modules that implement it (Part III).

A model, not a template. Other initiatives in the Carpathians — Ukrainian, Romanian, Polish, Slovak, Hungarian — work in different regulations, currencies, soils, and histories. What travels is the method, the sizing, the order of decisions, and the record of what failed. What does not travel is the answer. A guide that pretends otherwise is an export, and this work is not for export; it is with.

14. The unit: the module

Everything in the guide is a module with the same shape, so that a reader who has read one can navigate any other, and so that translation and revision stay tractable. The shape is fixed in the module template, in standards/.

Every module carries:

- **Purpose** — what problem it solves, in two sentences, and one line of connections: what the module depends on and what it feeds.
- **Pattern line** — under Connections: what it grows from and feeds among the place's own patterns, once their cards exist; which of Alexander's patterns it resonates with, as prompts and never as specification; and what it is in tension with (Section 27.3).
- **Context of validity** — the conditions under which the design holds: climate, altitude, group size, season of use. The most important section, and the one usually missing from published projects.
- **Sizing method** — the arithmetic, with the input a reader must measure themselves, not our result.
- **Bill of materials** — quantities, with substitutions that are known to work.
- **Build** — sequence, trades needed, time, what happens if it is done in the wrong order.
- **Cost** — with currency, month, and place of the quotation. A cost without a date is a rumour.
- **Operation and maintenance** — daily, seasonal, annual; who is responsible.
- **Failure modes** — what breaks, what it looks like when it is going wrong, what it costs to fix.
- **What we got wrong** — the section that makes the guide worth reading.
- **Regulatory notes** — by country, as a table, each entry dated and attributed to whoever checked it (Section 8.6).
- **Adaptation notes** — what a reader must change for a drier valley, a colder one, a smaller group; and the seed sentence: the design is a seed, not a copy — a different place grows a different form from it.
- **Provenance and licence** — who wrote it, from whose knowledge, under what terms, with its Gate 1 and wartime review.
- **What to check** — every figure marked [set] or [assumed], and how it will be replaced.

The three technical guides already drafted — water, energy, sanitation — feed twelve rows of the component map between them, not three. The map's source column shows which. Converting them is the first task of the pathway, not a later tidy-up.

15. Maturity: what may be claimed

Every module carries a level, stated at the top. Readers act on the level, not on the confidence of the prose.

Level	Meaning	May publish
L0	Idea, discussed	nothing, or a marked draft
L1	Designed and costed, basis per line	design, marked untested
L2	Built here	build record, real cost
L3	Measured through a full season	performance numbers
L4	Built by someone else, elsewhere	their figures beside ours
L5	Improved by others, merged back	the improved version

Whether a cost is quoted, estimated, or assumed is a label on each cost line, not a level. A module at L1 may be costed entirely from estimates, and says so line by line.

A marked draft. An L0 module that carries a sizing method and a design may be published as a marked draft: its status says so, its first lines say so, every figure in it is marked, and it carries Gate 1 and a wartime review like any other release. It claims nothing; a reader acts on L0. Decided 2026-09-28, so that work in progress can be read and co-created in the open.

The three existing guides sit at L1. Nothing in them should be read as tested, and the modules must say so on their face. The temptation the pathway has to resist is publishing L1 work in L3 language.

A module's maturity level says what is built. A pattern card's status says what is known about the place. The two are kept apart (Section 27.5).

16. The component map

Source says where existing material lies: W, E, S for the water, energy, and sanitation guides with their section numbers; P for the programme proposal, published among the grounding documents; 7a for the project's child-protection rules, held in its working knowledge; I for the infrastructure proposal, not yet located; F for Jean-Martin Fortier's *The Market Gardener*, 2014, on which the market garden and compost modules draw.

16.1 Ground and safety

Module	Status	Source
Shelter: standard, siting, certification	L0	E 9, W 2
Warning, communications, evacuation	L0	—
Fire protection and night lighting	L0	—
Medical room and quiet room	L0	—

16.2 Utilities

Module	Status	Source
Water: spring, cistern, treatment	L1	W 1–5, 8–9
Rainwater and nursery irrigation	L1	W 6, 8.5
Energy: PV, storage, generator	L1	E 1–4, 6–7, 9
Micro-hydro, if the stream allows	L0	E 5, 7.6
Sanitation: terra preta toilets	L1	S 1–3, 8.1, 9–10
Greywater, reed bed, pond	L1	S 6–7, 8.5–8.6
Wood heat and hot water	L0	E 8, S 8.7

16.3 Buildings

Module	Status	Source
Sleeping accommodation, winter-capable	L0	—
Kitchen and dining to institutional standard	L0	—
Washroom block	L1	S 8.7–8.8
Covered gathering place and hearth	L0	—
Workshop and tool store	L0	—
Insulation and mountain building practice	L0	—

16.4 Land, forest, food

Module	Status	Source
Tree nursery: beds, shade, watering	L1	I
Biochar: kiln, quality, storage	L1	S 2, 8.2
Terra preta: fermentation to soil	L1	S 4–5, 8.3–8.4
Planting, succession, the mountain calendar	L0	P 4.2
Compost: the garden stream to the beds	L0, marked draft	F
Market garden: beds, rotation, the season	L0, marked draft	F
Food storage: cellar, keeping, preserving	L0	—

16.5 Programme

Module	Status	Source
Pedagogy: the pathway and its stages	L1	P 3, 5
Rhythm: the day, the session, the year	L1	P 4, 5
Pattern cards and the record	L1	P 3.1
Elders' strand and its consent practice	L1	P 4.4
Safeguarding: the two tiers and the gates	L2	7a, P 6
Mentor preparation	L0	—
Adult strand: retreat and continuing education	L0	—

16.6 Organisation

Module	Status	Source
Legal form and ownership	L0	—
Budgeting: capital and running	L1	W 8, E 7, S 8, I
Funder fit and application practice	L1	P, I
Cross-border partnership	L0	—

Module	Status	Source
Documentation toolchain and standards	L2	standards/

16.7 Settlement

Module	Status	Source
Settlement form and siting, schematic	L0	—
Dwellings for resident households	L0	—
Paths, gateway, and access	L0	—
Common land and shared work	L0	—
Membership and self-governance	L0	—
Growers' and eaters' association	L0	—

Self-governance is designed threefold (Section 25). The growers' and eaters' association is its economic sphere, and the first part of it to be designed: the hamlet first, the region only once the hamlet is fed.

Forty-two modules. Two at a level where the site's own experience backs them, fifteen at design stage, two marked drafts at L0, twenty-three untouched. Naming the empty rows is more useful than filling them with prose.

Across all seven groups runs a pattern layer — the place's own patterns, discovered, not imposed — carried in the pattern record and in each module's pattern line. It is a second way of reading the map, not a group in it (Section 27.1).

Two L0 rows already have costed material in the guides — micro-hydro and wood heat — held at L0 because each waits on a measurement or a choice not yet made: the stream's head and flow, and the buildings the heat serves.

17. The pathway, in five phases

Phase 0 — Standards and repository. Fix the module template, the file naming, the versioning, the units and currency conventions, the image rules, and where the repository lives. Convert the three existing guides to module shape. Nothing is published yet. Weeks, not months, and it is the phase most often skipped.

Phase 1 — Document what exists. Before any new building: measure and write up what the site already has. The spring's yield. The existing flush toilets and where they discharge. The buildings, their construction, their heat loss. Walk the settlement with Alexander's patterns as prompts (Section 26) and begin the pattern record. This is the baseline every later figure is read against, and it can only be captured once.

Phase 2 — Build and measure. Each module moves L1 to L2 as it is built, with the build recorded as it happens — dated photographs, real invoices, the day the mason said the wall needed to be thicker. Then a full season of measurement before any performance claim: water in the dry month, battery through December, the reed bed under snow.

Phase 3 — Publish. English settles first; the Ukrainian sibling follows and is corrected over time rather than held back. Publication is module by module as each reaches L2, not a single release at the end. A module that is honest about being L1 can be published early and marked.

Phase 4 — Invite replication. One or two sites first, not a launch. A site that is already building something and needs two of our modules is worth more than fifty readers. Their figures come back into the module beside ours, which is what L4 means.

Phase 5 — Commons governance. Once other people's work is in the guide, it needs a way of deciding what gets merged, who holds the repository, and what happens if the original site stops. Deferred deliberately — do not build governance for a commons that has one member.

18. The first twelve months

A sequence, not a schedule. Each step assumes the one before. Section 3 says where the work stands.

1. Fix the module template, the repository, and the site parameters. Convert the three guides into the twelve rows they feed.
2. Baseline the site: spring yield in the dry month, existing sanitation and its outfall, buildings and their condition, the stream if there is one. Walk the settlement with Alexander's patterns as prompts, and begin the pattern record.
3. Send the written enquiries: establishment category, the spring's protection zone, discharge to ground, a fire-fighting reserve, a water-use permit for the stream. File each answer against the regulatory row of the module it governs.
4. Write the safeguarding module first among the programme modules — it is at L2 already and it governs the rest.
5. Publish the utility modules at L1, marked as untested, in English and Ukrainian.
6. Build the safety and utility layers; record as built, not after.
7. Measure one full season, including one winter.
8. Move the built modules to L2, then L3 where a season supports it.

9. Find one other site. Not a network, not a launch. One.

19. Licensing, and what cannot be licensed

Documents: CC BY-SA 4.0, as everything else in this project. Share, adapt, attribute, keep it open.

Designs and drawings: a hardware-appropriate licence rather than a document licence — CERN-OHL-S is the usual choice for open hardware that must stay open. The distinction matters the first time someone builds from a drawing and improves it.

Measurement data: released openly, ideally with no conditions beyond attribution, since data that comes with restrictions stops being compared.

Attribution is joint. The philosophy already carries two hands. The technical modules will carry more: the household, the mason, the electrician, the elders (Section 8.4).

Not ours to license. Hutsul practice, and the knowledge of any people, is not open because we wrote it down (Section 8.5).

20. How another site would use this

The replication protocol, written from the reader's side:

1. **Self-assessment.** Group size, season of use, altitude, water source, grid, legal status. Half a page, which decides which modules apply.
2. **Measure first.** Each module names what must be measured before it can be used. A site that skips the spring measurement and builds our cistern has not replicated anything.
3. **Take modules, not the model.** A site that needs only sanitation takes sanitation. Walk your own settlement with Alexander's patterns as prompts, and begin your own pattern record, as in Section 26. Your knowledge-holders name your patterns; ours are not yours.
4. **Localise the regulation.** Your row in the table, checked by you, sent back.
5. **Build, record, return figures.** Real costs in your currency and month; what you changed and why.
6. **Twinning, if wanted.** A working relationship between two households beats any amount of documentation, and the documentation exists mostly to make the first conversation shorter.

A failure log is part of the offer: sites that tried something from the guide and abandoned it, with the reason. The absence of such a log in most published projects is why most published projects cannot be trusted.

At another site, your own word replaces присілок; your own knowledge-holders stand where "the elders" stand here; and your own law replaces every Ukrainian reference in this document.

21. Who pays for the documentation

Documentation time is real work — roughly a day per module to draft, and more to check, draw, and translate. It is invisible in most budgets, which is why most projects do not have documentation.

Three routes, not exclusive:

- **Inside the infrastructure budget.** A documentation line of five to eight percent of capital, justified as open educational resources, which several funders fund willingly.
- **As its own small grant.** Open knowledge, digital commons, and educational resource programmes fund exactly this, and the deliverable is unusually easy to verify.
- **In kind.** The household's time, counted as contribution at the same rate as toloka labour, so its value appears rather than vanishing.

The guide is not sold and is not a product. If someone wants more than the guide gives — a person on site, a design for their valley — that is a separate arrangement and does not close the guide.

Part III — The open hamlet: how the settlement grows and governs itself

22. The question

Is the forest school, rightly seen, a hamlet — a small settlement of households, work, learning, and land — run as a living lab, and published as an open hamlet the way the modules are published as open educational resources?

The forest school is not only a place for children. It is also a place for adults — retreat and continuing education — and around it a small settlement is planned: households living there, with work, learning, and land held together. The open hamlet is that settlement, documented as openly as the modules. It is run as a living lab, and it comes to know itself through citizen science (Section 9.5).

The answer taken on 2026-09-27 was yes, subject to the checks with the elders (Section 29), with three consequences for the component map. Programme gains an adult strand — retreat and continuing education. Legal form and ownership stays an L0 row and becomes a precondition: nobody lives on the site before it is settled. Safeguarding must cover retreat guests and residents, not only visiting adults during sessions: a design question for the hamlet.

23. What a hamlet is in this landscape

23.1 As the literature holds it

Every point below is for the elders to confirm, correct, or withhold before it is written down as ours [for the elders].

- **Dispersed settlement.** Separate homesteads scattered on the slopes, far from each other, placed near the pastures rather than gathered in the valley.
- **Village core and outliers.** A core in the valley — church, school, shop — with homesteads standing singly or grouped in outlying parts called kut, which belong to the village but lie far from it.
- **The enclosed homestead, гражда (grazhda).** House and farm buildings joined by a high roofed wall into a closed yard with one gate; described as the form of the better-off, standing beside the open yard with freely placed buildings, and now seen mostly in museums.
- **Homestead, hayfield, forest, polonyna.** A land mostly of forest and meadow, with little arable ground; summer pasture on the polonyna, with its own seasonal buildings and enclosures, and the herd back down by the end of September.
- **толока (toloka).** Used in the programme for shared work; recorded in one study of the region also as family pasture held across generations near the homes (Section 10).

The literature disagrees on the altitude of traditional dwellings. No figure enters our text until it is resolved.

23.2 The word for it

Word	What it carries	Use
присілок	small settlement belonging to a village	working term
кут	Hutsul outlying group of homesteads	for the elders
поселення	settlement; since 2024 also a legal term	avoid loosely
спільнота	the community of people	for the people
громада	also the territorial community	avoid
хутір	isolated farmstead	too small

Working title in Ukrainian: відкритий присілок. The elders' word replaces it if they give one.

A legal note, unchecked. Law of Ukraine No. 3285-IX, in force from 26 January 2024, recognises town, selyshche, and village as the categories of populated place, and is reported to give поселення a narrow legal sense: a compact place of residence outside any populated place, without a stable

population. Read from secondary sources only; the statute text is to be read before the word is used. A seasonal retreat and education site may fall inside that sense, which makes the word a legal question and not only a translation.

One term avoided. Родове помістя and the settlements built around it belong to a particular movement, and a plain екопоселення carries some of that association.

24. Two languages, kept apart

Two questions are asked of the settlement and kept apart (Section 9.2):

- which of Alexander's patterns a forest school hamlet already carries or needs — Section 26;
- which patterns only this place, its household, and its elders can name — the discovery work, recorded on pattern cards, never pre-filled here.

Where an Alexander pattern and a likely local form pull against each other, the tension is named and left for the place to answer.

25. Three spheres, as a lens

The hamlet's adults govern themselves with a lens taken from Rudolf Steiner's social threefolding (Section 10), used as method and not as worldview — never taught to children, and asked of no family. Three questions are put to every decision:

- **Cultural life — whose freedom is this?** Learning, the adult strand, the elders' culture, each family's faith, the pattern record: run by those who carry them, and funded by gift, never by sale. Where teaching is paid, as in retreats, the price is set in the economic sphere and the content stays with those who teach.
- **Rights life — what holds for everyone alike?** Standing rules, membership, consent, the tenure of the land, and safeguarding above all: equal for every person, and not for sale to money or pedagogy.
- **Economic life — what need, met by which association?** Meals, trades, shared work, the nursery, the garden, retreat income: agreed among those who produce and those who use. The garden's association of growers and eaters has two circles — the hamlet first, the region only once the hamlet is fed.

Self-governance, designed threefold. Those who teach decide pedagogy; all members decide the standing rules, each with one voice; the associations of those who produce and use decide the economy. Safeguarding sits in the rights sphere, beyond the reach of the other two.

The lens is not taken one to one. The theory is contested; the state is part of the rights sphere, and under martial law more than ever; and the place may hold its own forms of the same three, or others, for the elders to name.

Where the same few people sit in all three spheres — as they will in a hamlet of a few households — the lens still asks which question is being decided, not who decides it.

A caution from practice. Camphill communities, partly organised on these lines, lived for decades with co-workers who were unpaid and housed as families. At one of them, from 2014, that model met charity governance and employment rules, co-workers began to be made employees, and residents' legal challenge was refused in 2015. No economy here is designed to rest on unpaid residents without a footing in Ukrainian labour law and the rules for children's establishments.

Where the spheres show in Alexander's prompts.

- **Cultural:** 18 Network of Learning, 43 University as a Marketplace, 83 Master and Apprentices, 85 Shopfront Schools, 181 The Fire, 185 Sitting Circle.
- **Rights:** 36 Degrees of Publicness, 44 Local Town Hall, 53 Main Gateways, 61 Small Public Squares, 67 Common Land, 69 Public Outdoor Room.
- **Economic:** 9 Scattered Work, 41 Work Community, 46 Market of Many Shops, 87 Individually Owned Shops, 91 Traveler's Inn, 148 Small Work Groups, 156 Settled Work, 157 Home Workshop, 177 Vegetable Garden, 178 Compost.
- **Where they meet, and where one most easily swallows another:** 45 Necklace of Community Projects, 80 Self-Governing Workshops and Offices, 129 Common Areas at the Heart, 147 Communal Eating.

Patterns named here but not in Section 26 have not been walked; they are for the reader to look up.

26. Alexander's patterns — prompts for the settlement walk

26.1 How to read this list

It is a set of prompts for the Phase 1 walk of the settlement, used beside the baseline. Patterns are cited by number and name; the descriptions are ours.

Walk status — distinct from a module's maturity level and from a pattern card's status — is one of four:

- **present** — the loaded documents record it on the site;
- **absent** — the documents record that it is missing;
- **unknown** — nobody has looked yet;
- **planned** — a module at L1 already designs it.

A commitment in the programme proposal is not a record of the site. It is noted beside an unknown, never counted as present.

Most rows are unknown. That is the finding: the baseline has not been walked.

Absent is a description, never a verdict. A pattern this place does not carry is a question — what answers that tension here? — not a gap to fill (Section 9.2).

Sacred sites, holy ground, and grave sites — 24, 66, 70 — are recorded nowhere, in workshop or publication, unless the elders ask.

26.2 Community and neighbourhood

- **7 The Countryside** — farmland and forest open to people on foot, as a park is. Unknown. Tells us: tenure and rights of way; elders.
- **18 Network of Learning** — learning spread through working life, not held in one building. Unknown; the programme proposal commits to it. Tells us: whether learning happens in workshop, kitchen, and forest.
- **24 Sacred Sites** — not recorded; see 26.1.
- **25 Access to Water** — a belt of common land kept along natural water, with settlement reaching it only here and there. Unknown. Tells us: the stream's banks, if there is a stream, and the Water Code's protective strip, which keeps the compost bays and the beds out of that belt.
- **35 Household Mix** — different households side by side. Absent; the hamlet is planned. Tells us: who the hamlet is for.
- **40 Old People Everywhere** — the old woven in, not set apart. Unknown; elders visit, none live here. Tells us: whether the hamlet makes room for elders to live.
- **41 Work Community** — work places as a small community near homes. Unknown. Tells us: whether workshop, nursery, and kitchen form one.
- **45 Necklace of Community Projects** — small projects at the edge where the public meets them. Unknown. Tells us: whether there is a threshold where the valley comes in.
- **53 Main Gateways** — a marked entry. Unknown. Tells us: the walk; the gate is also a safeguarding line.
- **67 Common Land** — land held between households. Unknown. Tells us: the elders; see *толока* in 23.1.
- **74 Animals** — animals kept among people. Unknown. Tells us: a household decision; whether the garden stream's manure is made or bought.

26.3 Cluster of buildings

- **36 Degrees of Publicness** — dwellings from open to secluded. Unknown. Tells us: the plan; guests, residents, and children need different degrees.
- **37 House Cluster** — houses around shared ground. Absent. Tells us: the plan; in tension with dispersal, 26.6.
- **83 Master and Apprentices** — the skilled and the learning side by side. Unknown; the programme proposal commits to the elders' sessions. Tells us: whether the workshop layout allows it.

- **91 Traveler's Inn** — a place to stay where those passing through form a company of their own for the evening. Absent; the adult strand is planned.
- **95 Building Complex** — a programme in several small buildings. Unknown. Tells us: the Phase 1 building inventory.
- **98 Circulation Realms** — the site legible as realms in sequence. Unknown. Tells us: the walk.
- **104 Site Repair** — build on the worst ground, keep the best. Unknown. Tells us: every siting decision.
- **105 South Facing Outdoors** — buildings north of their outdoor space. Unknown. Tells us: the walk; the enclosed homestead in the literature puts the house on the north side of its yard.
- **115 Courtyards Which Live** — enclosed yards that are used. Unknown. Tells us: the grazhda question, for the elders.
- **147 Communal Eating** — one table at the heart. Unknown; the programme proposal commits to full meal provision. Tells us: the kitchen and dining module, L0.

26.4 The building

- **47 Health Center** — health care built around the activities that keep people well. Unknown. Tells us: the medical room module, L0.
- **127 Intimacy Gradient** — rooms from public to private. Unknown. Tells us: each plan; safeguarding reads it too.
- **129 Common Areas at the Heart** — shared rooms where paths pass. Unknown. Tells us: the main building plan.
- **137 Children's Realm** — space shaped for children within common life. Unknown. Standing design in the plan only.
- **139 Farmhouse Kitchen** — cooking and eating in one room. Unknown. Tells us: the kitchen module; institutional rules may forbid it.
- **154 Teenager's Cottage** — a room of one's own, at some distance. Unknown. Standing design only.
- **155 Old Age Cottage** — a small dwelling for the old, near others. Absent. Tells us: the hamlet design.
- **156 Settled Work, 157 Home Workshop** — work within reach of living. Unknown. Tells us: the workshop module, L0.
- **181 The Fire** — a real fire at the centre. Unknown. Tells us: the gathering place and hearth module, L0; wood heat in the energy guide.
- **185 Sitting Circle** — seats in a round. Unknown; the programme proposal commits to a daily circle. Tells us: whether a room holds it in bad weather.
- **186 Communal Sleeping** — shared sleeping with privacy within it. Unknown. Tells us: the sleeping module; safeguarding sets the form.

- **197 Thick Walls** — walls deep enough to hold shelves, seats, and niches. Unknown. Tells us: the building survey.
- **203 Child Caves** — small hollows. Unknown. Standing design only.

26.5 The land between

- **62 High Places** — a point from which the whole is seen. Unknown.
- **64 Pools and Streams** — surface water left running. Unknown. Tells us: the stream; the rainwater module.
- **71 Still Water** — still water for swimming, entered from a shallow edge. Unknown. Tells us: the pond in the greywater module is a treatment pond, not a swimming pond — a tension to name there.
- **106 Positive Outdoor Space** — outdoor space shaped as rooms. Unknown.
- **114 Hierarchy of Open Space** — every outdoor place with a back and a view. Unknown.
- **120 Paths and Goals** — paths where feet already go. Unknown. Tells us: worn paths, read as the toolkit reads wear.
- **168 Connection to the Earth** — buildings meeting the ground softly. Unknown.
- **169 Terraced Slope** — slopes worked in terraces. Unknown. Tells us: the elders.
- **170 Fruit Trees** — fruit trees near dwellings. Unknown.
- **172 Garden Growing Wild** — gardens partly left to themselves. Unknown; the programme proposal's regeneration strand points here.
- **177 Vegetable Garden** — ground set aside for vegetables, sunny and central to the households it serves. Unknown; the market garden module, L0. Tells us: where the deepest, sunniest ground lies, and whether it is hayfield.
- **178 Compost** — toilets and kitchen waste composted together for the land. Planned for the toilet stream — terra preta, L1, to trees only. Unknown for the garden's own stream, kept apart from it: the compost module, L0, a marked draft. Tells us: the market garden and compost modules.

26.6 Where the tension bites

1. **Cluster against dispersal.** 37 House Cluster and 35 Household Mix assume a face-to-face group of dwellings. The literature holds the Hutsul form as dispersed homesteads, grouped at most as a kut. The place may hold a different answer — dispersed households joined by paths and shared work.
2. **Gradient against enclosure.** 127 Intimacy Gradient and 36 Degrees of Publicness assume a soft gradient; the grazhda is a hard wall with one gate. The literature also reports, in a few homesteads, a kept front yard and a working back yard inside it — a gradient within the enclosure, and a candidate pattern for the elders to name or not.
3. **Valley against slope.** 4 Agricultural Valleys protects valley farmland. Here the counterpart may be the hayfield. For discovery.

4. **Common land and toloka.** The one study cited records toloka as family-held pasture, not common land — it may pull against 67 Common Land rather than match it. Whether anything here is held in common is for the elders.
 5. **Sacred sites and graves.** Patterns Alexander would have us protect by rule and by plan. Here they enter no plan unless the elders ask.
-

27. The bridge between patterns and modules

27.1 What is true

Modules are not simply the construction-scale implementation of patterns, and patterns are not a layer above the component map. The relation is many to many: one module touches several patterns, and one pattern draws on several modules. Patterns cut across the map as a second way of reading the same place — the map says what is built, the patterns say why and where it belongs.

Evidence flows back. A module measured through a season is citizen science data for the pattern cards it touches. Patterns give modules their reason; modules give patterns their evidence.

27.2 Three kinds of pattern

Kind	Named by	Cited as
The place's	household, co-creators, elders	card name
Alexander's	the book, as a prompt	number and name
A module's design rules	us, as designers	the module's text

A module's own design rules — the water module's four rules — stay inside the module and are tested by its maturity level, not by pattern status.

27.3 The pattern line

Every module carries, under its Connections line:

- **Grows from** and **Feeds** — the place's patterns, by card name, once the card exists; until then, "not yet named";
- **Resonates with** — Alexander's, by number and name, as prompts;
- **In tension with** — any pattern the module pulls against.

Only the place's patterns are grown from and fed. Alexander's are resonance, never specification. The project's adapted pattern card carries the return line, **Implemented by**, naming the modules. The Erdpuls card is not edited.

27.4 The water module as the example

- **Grows from:** not yet named. Two questions for the elders before the spring box is built — how homesteads here stand to their spring, and whether this spring is held in any way that decides how it may be touched. The SPADOK heritage page on the Hutsul dwelling places the outbuildings of a yard by relief, road, garden, and distance to the spring: a candidate place pattern, the elders' to name.
- **Resonates with:** 207 Good Materials, 83 Master and Apprentices, 104 Site Repair.
- **In tension with:** 64 Pools and Streams — the buried cistern and piped main take water out of sight; the spring and catchment are where it can stay visible. 25 Access to Water — the catchment fence keeps people back from water the pattern would open to them.

27.5 Status and level

Pattern status — first observation, confirmed, contested, refined, archived — says what is known about the place. Maturity level says what is built. They meet in one place only: a module at L3 is a data source for its patterns' cards. No card is confirmed on a module's evidence alone; it still needs independent observations across groups and seasons.

28. The open hamlet

28.1 What the open hamlet is

Two things published side by side, neither complete without the other:

- **The pattern language of this place** — discovered, not imposed; named by the household, the co-creators, and the elders; recorded on cards; told once a year as a narrative, as the Pattern Language Assembly Guide sets out.
- **The modules** — how the place is built and run, each carrying its pattern line.

The hamlet itself is never published as a plan. What travels is the language and the modules: a seed, not a copy.

Published, when each is ready: modules from L1, marked; the pattern language as a narrative, at valley resolution, after the year's synthesis; schematic diagrams of relationship between parts of the hamlet.

Not published: any site plan with distances, until the decision on naming the site; pattern cards in full, which carry the most detail of place; anything of the elders' without their consent, or of a resident household's life without its own Gate 1 (Section 2).

28.2 The pattern card, adapted

The Erdpuls card, with six rules for this project:

- no child's name; discoveries attributed to the cohort;
- place recorded at ring level only — body, homestead, garden and forest edge, village, valley — never by coordinates;

- names in the discoverers' own language first;
- an **Implemented by** line naming the modules;
- an open **Form** field — pattern, rhythm, movement, relation, or another the place shows — so the card does not force Alexander's shape on what is found;
- an optional **Sphere** field — cultural, rights, economic, or where they meet — left empty when none fits.

The Erdpuls card lives in the Pattern Discovery Toolkit, EN v1.3, in the Erdpuls OER collection. The adapted card is standards/template_pattern_card_EN_v0_1.md. Pattern files in the workshop are named pattern_[name][*LANG*]_v[*major*][*minor*].md.

29. To confirm with the elders

- The local names: kut, присілок, осідок, гражда — and whether any names what is being built.
 - Whether this valley had enclosed or open yards.
 - The valley's own relations of homestead, hayfield, forest, and polonyna, and their seasons.
 - Which sense of toloka is used here.
 - Whether the boundaries of the place are theirs to draw, and how.
 - Whether any of this may enter the guide at all, and on what terms.
-

Part IV — Decisions and record

30. What has to be decided

- Whether the guide names the site at all, ever — and if so, at what point in the war.
 - Which hardware licence, decided before the first drawing rather than after.
 - Who the co-authors are, named on each module, and how the elders' material is handled.
 - Whether documentation is funded inside the infrastructure budget or separately.
 - Which country a partner might come from first, since that decides the next language.
 - How much of the programme is published.
 - Legal form and ownership, before anyone lives on the site. One option to weigh, not decided: land held in trust, out of the market.
 - Each resident household's own Gate 1, before its life is described.
 - How safeguarding supervision extends to retreat guests and residents.
 - How much of the adult strand is published.
-

31. Decisions taken

2026-09-27

- **The repository:** a private workshop on GitHub under ubecccommon; publication as a collection in ubecccommon.github.io (Section 3).
- **Grounding documents:** the programme proposal, the philosophy, and the comparative survey are published as working drafts, in standing format, in foundations/EN/grounding/.
- **Names:** released modules carry the author's name and the initiative's name.
- **Figures:** capacity, session length, and use days may be published, as standing format and never beside a date.
- **Living lab:** a descriptive frame only; no connection with any network or certification body.
- **Participants:** co-creators, children and adults alike; never users.
- **Citizen science:** woven in, under the valley-resolution rule.
- **The adult strand and the settlement:** part of the concept.
- **Working term:** присілок, pending the elders.
- **Alexander's list:** prompts for the Phase 1 settlement walk.
- **Walk status:** present, absent, unknown, planned; a proposal's commitment is noted beside unknown, never counted as present.
- **Sacred sites, holy ground, grave sites:** recorded nowhere unless the elders ask.
- **Not one to one:** no pattern language is taken over whole; Alexander's patterns are prompts, and the place may show forms that are not patterns in his sense.
- **Thinking like a plant:** the hamlet grows piecemeal, from what is there, in season, as a seed and not a copy.
- **The bridge:** three kinds of pattern; the pattern line in every module — grows from, feeds, resonates with, in tension with — added to the template and the water module.
- **Three spheres:** social threefolding as a lens for the adults' self-governance, never a doctrine; an optional Sphere field on the adapted card; the self-governance module designed threefold.
- **The open hamlet:** published as the place's pattern language and the modules, never as a plan; hamlet-scale plans as relationships only, without distances, until the site-naming decision; pattern cards kept in the workshop, only the yearly narrative released.
- **The map:** a seventh group, settlement; a pattern layer across all groups; the adult strand added to programme. Forty modules.
- **Legal form:** land held in trust, out of the market, named as one option to weigh; not decided.

2026-09-28

- **Languages:** the published foundations, the grounding documents, the water module, and the two templates are published in Ukrainian and German, as machine translations to be reviewed; English authoritative.
- **The compendium:** the foundations, pathway, and open hamlet papers gathered into one document, with an overview of the project, its structure, and its documents, and published beside them; the three stay published.
- **The market garden:** the kitchen garden row replaced by a market garden and a food storage row in Land, forest, food, and a growers' and eaters' association in Settlement; one association, two circles, the region opened only by a measured surplus. Forty-two modules.
- **Two nutrient streams:** the toilet stream to trees only, as the sanitation paper holds; the garden fed from its own stream.
- **The garden's footing:** a registered capacity of the operator that runs the kitchen; preserves made for the hamlet only; raw vegetables to the hamlet and, once it is fed, to the region.
- **The budget share:** both circles carry the garden's budget by pledge, as at Temple-Wilton and in the Solawi bidding round; the second circle shares only the risk of the surplus, and its terms say so.
- **The name:** market garden, the name of the method; nothing in it is sold at a market.
- **A marked draft:** an L0 module with a sizing method and a design may be published as a marked draft, with Gate 1 and a wartime review; the market garden module is the first.
- **The compendium with every module:** each time a module is published, the compendium is brought up to date in the same release — the module's row in Section 16, the counts in Sections 1 and 16, where the work stands in Section 3, the published modules in Section 4, and a line here — and released with it.
- **Ecologically oriented:** the market garden's practice is called ecologically oriented, not organic; no certification is sought for now, and no produce is labelled organic, bio, or eco.
- **Compost, two rows:** the compost row of the map is the garden stream — kitchen, garden, and manure to the beds — and the terra preta row carries the whole toilet stream, fermentation, worms, and maturation, to trees only; one kiln feeds both, raw char at the seat and charged char in the pile; manure is an input, not a module, until the household keeps animals; woody clearings go to the kiln and forest clearings stay in the forest. Cooked food, plate waste, and grease-trap solids go to the toilet stream's vessels, to trees; the garden stream takes raw plant material, residues, charged char, and manure only. Compost never leaves the hamlet. Leaves come from the yard and the fruit trees, never raked from the forest, until the elders speak. The Water Code's protective strip, 50 m on a slope, keeps bays, piles, and beds back from the stream and the pond; the market garden is brought into line in its next version.

2026-09-29

- **The compost module:** published at L0 as a marked draft, v0.1, in English, Ukrainian, and German — sized from the market garden's need, with the beds built from bought rotted manure and 11 to 21 m³ bought every year after, priced from a dated listing; its distances are to be measured on the walk.
 - **Twelve rows:** the three working-paper guides feed twelve rows of the map, not eleven, now that the compost row no longer draws on the sanitation guide (Sections 14 and 18).
 - **One toilet stream:** nothing is separated. Urine is not diverted, as the sanitation paper holds; the buckets and the kitchen's cooked food, plate waste, and grease-trap solids go into the same sealed vessels. The terra preta module takes what comes and is sized for it, with a smaller case if plumbed units are required for the children's side.
 - **The toilet stream, as designed:** a carbon cover — sawdust, dry leaves, or shredded straw — at the seat beside the char, since char alone cannot hold the urine; the worm beds fed fermented material layered with bulking and finished vermicompost, with grease-trap solids a small share; each bay filled as a batch and closed, its year counted from the last addition; use on trees, hedgerows, windbreaks, and nursery stock only, and nothing in the month before a fruit harvest. No vessel, store, bed, or bay is bought or built before a written answer from the sanitary service on MOH Order No. 145, point 2.21.
 - **The kiln:** a flame-curtain cone of the Kon-Tiki type, welded locally from 3 mm steel to the published design, with a soil pit as the alternative; dry wood only, at the fire distances, quenched from the pond or rainwater, never the spring; no certificate claimed.
 - **Three modules published:** terra preta, v0.1, and biochar, v0.1, at L1, converted from the sanitation paper and recalculated at capacity, and the market garden at v0.4, with the compost priced and its beds, bays, and heap kept out of the Water Code's protective strip; in English, Ukrainian, and German, with their PDFs.
-

32. Reviews and stress tests

The open hamlet, 2026-09-27. Three independent reviews, each by a reader who had not seen the work being produced.

- **Safeguarding and aggregation.** A hostile reader with the workshop in hand. Found: a path by which published material could be joined to other writing about the place; a tool in the workshop carried material from that writing; the capacity figure sits in a releasable module while its publication is undecided; use days and session length together give the programme's calendar; the old plan rule, relationships and distances, stood unamended in four files; the release list refused too little. All corrected or raised as open decisions; the name and the figures were then decided for publication, the tool split, and a release script written.

- **Replication.** A small mountain initiative in another country trying to use the method. Found: the safeguarding tiers it rests on sit in a document it will never see; counts and references that disagreed between files; folders described as existing that do not; one word, status, doing three jobs; a paid retreat with no home among the three spheres; no step for its own settlement walk. All corrected.
- **Evidence.** Every claim checked against sources. Found: six of the Alexander descriptions misstated their pattern — 7, 25, 47, 71, 91, 197 — and the water module's tension was named with the wrong pattern; toloka as family pasture pulls against common land rather than matching it; statuses of present and planned rested on a funding proposal or on L0 rows. All corrected. Checked and found right: every pattern number and name, the forty-module count, the Botton Village dates, the law's dates, the altitude disagreement.

The foundations paper, 2026-09-27. An independent review before release. Found and corrected: the Berg and Dasmann sentence cited to the reprint in which it appears; the Erdpuls method claimed as designed and documented, and no more; the module counts; the wording on naming the site, which holds while the decision is open.

The comparative survey, 2026-09-27, was source-checked and stress-tested on its own; it records what was found.

33. Where this document came from

Gathered on 2026-09-28 from three papers, which stay published as working drafts:

- the foundations paper, EN v0.3 — Parts 0 and I;
- the pathway paper, EN v0.4 — Part II and Section 2;
- the open hamlet paper, EN v0.1 — Part III, and Sections 9.2, 9.4, and 9.5.

What changed in the gathering:

- **New:** Sections 3, 4, and 5 — how the work is organised, the documents, how to read this document — and Section 3's account of where the work stands.
- **Safeguarding** moved to the front, as Section 2, joined by the open hamlet's rules on what is published, and by the rule held in the project's working instructions: when in doubt, the hold closes.
- **Joined where the papers repeated each other:** Open Source Ecology; not ours to give; across the borders; growing like a plant; citizen science and the living lab; the measure of completion; social threefolding and the Camphill caution. The fuller text was kept in each place, and nothing was dropped.
- **Moved:** the toolchain and the translation policy into Section 3; the pattern line into the module's shape; the open decisions into Section 30 and the decisions taken into Section 31, with their dates.
- **Added:** Section 29 asks the elders about the boundaries of the place, which Section 8.6 marks for them.

- **Dropped as done:** the open hamlet paper's lists of changes to be made to the pathway paper, the template, and the repository, all applied on 2026-09-27.
- **Section numbers** differ from the papers'. Documents that cite a paper keep citing it, since the papers stay published: Section 11 of the pathway paper is Section 2 here; the foundations paper's Sections 3, 6, and 8 are Sections 7, 10, and 12; the open hamlet paper's Section 6.3 is Section 28.2; the pathway paper's Sections 4 and 5 are Sections 15 and 16.

The three papers stay published, in all three languages, beside the compendium. A change made in one is carried to the other; where they differ, the compendium holds.

v0.2, 2026-09-28. The market garden enters the map (Sections 16.4 and 16.7), with the counts in Sections 1 and 16, the economic sphere in Section 25, the walk of 177 and 178 in Section 26.5, and the decisions in Section 31; the compost row made the garden stream, and the terra preta row the whole toilet stream (Sections 1, 16, 26.2, 26.5, 31). On 2026-09-29 the compost module was published beside the market garden, and the count of rows the three guides feed corrected to twelve (Sections 1, 3, 4, 14, 16, 18, 26.5, 31, 34).

v0.3, 2026-09-29. The toilet stream decided as one stream, and the toilet stream and kiln as designed (Section 31); the terra preta and biochar modules published at L1, and the market garden at v0.4 (Sections 3, 4, 31, 34). The pathway and open hamlet papers still carry the forty-module map; they are brought into line in their next versions, and until then the compendium holds.

34. Release record

The compendium is released with the guide, from foundations/EN/, as a working draft, beside the three papers it gathers. It is changed as the work goes on, in the open, and each new version is released the same way, with both lines below dated again.

- **Gate 1:** household decision, 2026-09-29, recorded by Michel Garand — v0.3 may be released as it stands, as a working draft.
- **Wartime review:** 2026-09-29, Michel Garand — v0.3 read against Section 2; names no place; the new decisions name no market, no town, no buyer, no supplier, and no day; every site plan relationships only; accepted.

Version 0.2 was released on 2026-09-29 with the household's Gate 1 and a wartime review, both recorded by Michel Garand.

Version 0.1 was released on 2026-09-28 with the household's Gate 1, recorded by Michel Garand, and a wartime review by Michel Garand: names no place; every site plan relationships only; checked against the list of place names; accepted.

35. Sources

Searched 2026-09-27; to be re-checked before any release.

- Alexander, C., Ishikawa, S., Silverstein, M. et al., *A Pattern Language*, 1977 — pattern numbers and names, p. x and p. xv, pages to be checked against the print edition; the pattern descriptions here are ours, checked against the pattern summaries at patternlanguage.cc.
- Alexander, C. et al., *The Oregon Experiment*, 1975, on piecemeal growth.
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- Encyclopedia of Ukraine, entry Hutsul region, on settlement form and land use.
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- Irwin, A., *Citizen Science: A Study of People, Expertise and Sustainable Development*, 1995.
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- Open Source Ecology, the Global Village Construction Set.
- Ostrom, E., *Governing the Commons*, 1990.

- Past and Present of Hutsulshchyna as the Carpathian Borderland Region, on the enclosed homestead and its yards.
- Pestalozzi, J. H., head, heart, and hand, as summarised by the Pestalozzi foundations and by Arthur Brühlmeier, Head, Heart and Hand.
- Social threefolding and associative economics, as summarised in the Wikipedia entries of those names; Steiner's comparison with the threefold human organism, as summarised on rudolfsteinerweb.
- Steiner, R., the memoranda of 1917 and the threefolding campaign of 1919, with their critics, as summarised by the Institute for Social Ecology.
- Talka, tloka, toloka: the Latvian register of intangible heritage; Brückner, as cited on tloka; the 1557 Lithuanian estate regulations, as cited in Orbis Lituaniae.
- Third Sector, 2015, on the Botton Village judicial review.
- UNESCO, Recommendation on Open Educational Resources, 2019.
- Ukrainian sources on гражда and the polonyna holding: Wikipedia (uk), Гражда; the SPADOK heritage site, Гуцульське житло; a study of traditional Hutsul occupations on ResearchGate.
- Wikipedia, Ukrainian khata, on the grazhda among the better-off; Green Ukraine, on the kuty and on surviving grazhdas; Insights People, on the polonyna season ending in late September.
- Erdpuls OER collection: Pattern Discovery Toolkit, EN v1.3, and Pattern Language Assembly Guide, EN v1.2, 2026.
- This project: the grounding documents (Section 4); the project prompt, v0.2; the repository conventions, v0.5.

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